

Deep Learning–Based Markerless Motion Tracking for Canine Gait Analysis & CCL Injury Detection

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The Clinical Problem



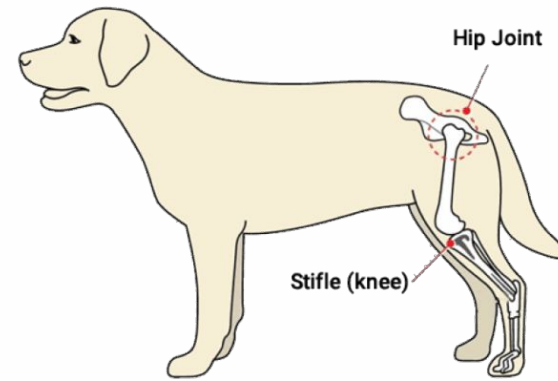
Cranial Cruciate Ligament (CCL) tears are a common and debilitating injury in dogs.



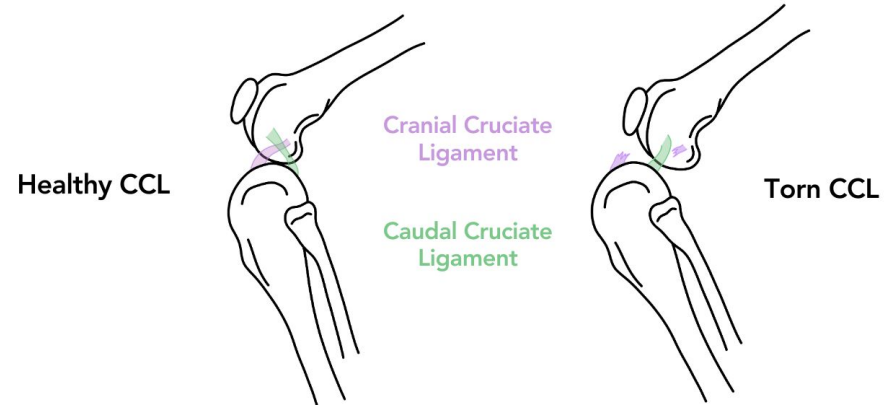
Assessing lameness is important for diagnosis and future treatment decisions.



Problem: How can we classify a dog's gait as healthy or injured outside of a specialized lab?



Inside the Canine Stifle (Knee) Joint



Limitations of the “Gold Standard”

Marker-Based Motion Capture

- Standard for kinematics, used to compute joint angles
- Cons: requires precise marker placement, expensive high-speed cameras, and special laboratory, hence impractical for routine clinical use

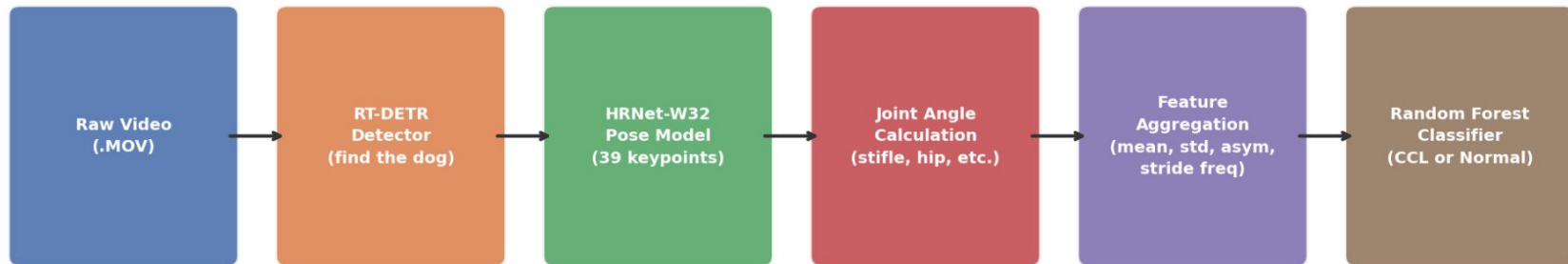
Pressure Sensing Walkways

- Standard for kinetics, used to measure ground forces
- Cons: expensive, requires specialized equipment, cannot capture joint angles or kinematic data, only forces applied by the paws



Solution: Markerless Gait Tracking Pipeline

Objective: We wanted to use gait features, especially left/right joint-angle asymmetry, to train a binary classifier to distinguish CCL-injured from healthy dogs



Dog Detection & Video Filtering

- Real Time Detection Transformer (RT-DETR) detects the dog and returns a bounding box per sampled frame
- The code tracks the center point of that bounding box across ~12 sampled frames
- It then measures how much the center moved horizontally vs vertically
- If horizontal movement dominates (ratio ≥ 2.5) and the dog traveled at least 20% of the frame width $\rightarrow$ classified as lateral
 - This helps pare down the massive dataset into usable, sagittal plane views of the dog walking, which carries all the kinematic information



Good



Not Good



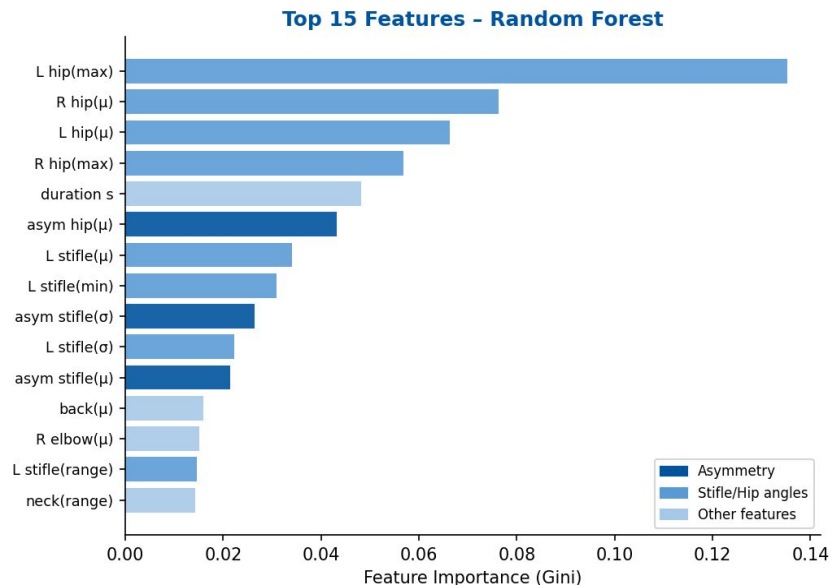
Pose Estimation

- **Model:** SuperAnimal-Quadruped, which is based on the DeepLabCut (DLC) pose estimation framework
 - Localizes 39 anatomical keypoints, including nose, ears, spine, shoulders, elbows, wrists, hips, stifles (knees), and paws on both sides of the dog per frame
- **Joint angles:** Stifle (knee), hip, elbow, and shoulder angles are computed from keypoint triplets each frame.
 - Left and right sides are tracked independently to measure asymmetry over the stride cycle.



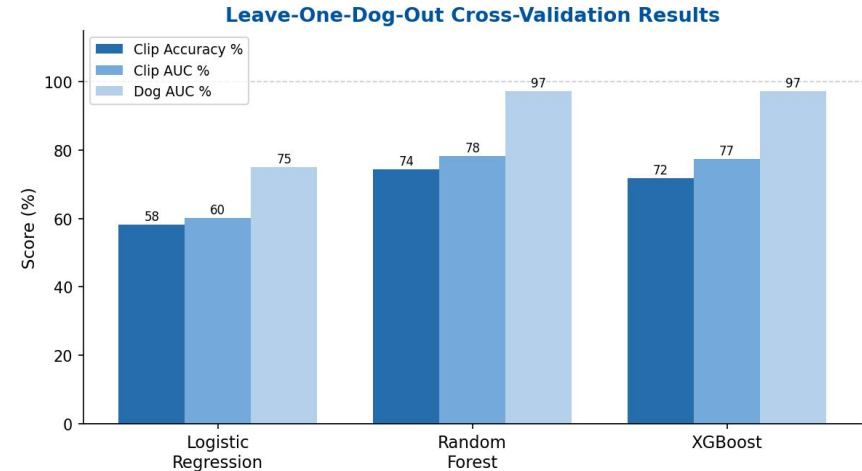
Feature Extraction

- **Angle stats (50):** Mean, std, min, max, range for 10 joint angles per clip.
- **L/R asymmetry (8):** Mean and std of $|L - R|$ for stifle, hip, elbow, shoulder.
 - Plausible manifestation of CCL injury
- **Stride & stance (7):** Stride frequency, stance variance ratio, duration, pose coverage, direction.



Model Training & Selection

- **Three candidates:** Logistic Regression, Random Forest, and XGBoost
- Model trains on 11 dogs while one is held out for testing, repeating 12 times until every dog has been the test dog exactly once.
- Dog level AUC was the most important metric, as it addresses the clinically meaningful question of whether or not the model correctly predicts a higher score for the injured dog



Results

- **Cross-validation:** Random Forest achieved a dog-level AUC of 0.972, XGBoost tied at 0.972, and Logistic Regression was last at 0.75
- **Holdout test (2 dogs, 56 clips):** Clip accuracy of 94.6% and AUC of 0.975. The 3 misclassified clips were all CCL clips called normal
- **Dog-level results:** when aggregating clip probabilities per dog, both Reggie Bell (CCL) and Zane Kleva (normal) were correctly diagnosed, demonstrating the model generalizes to dogs it has never seen

Holdout Confusion Matrix (Clip-level)

True: Normal	38 True Neg (Normal→Normal)	0 False Pos (Normal→CCL)
True: CCL	3 False Neg (CCL→Normal)	15 True Pos (CCL→CCL)
	Pred: Normal	Pred: CCL

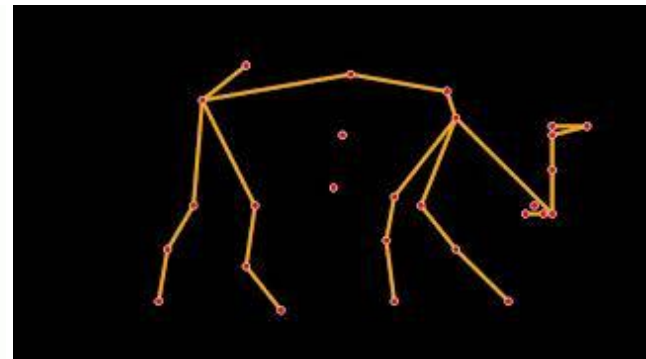
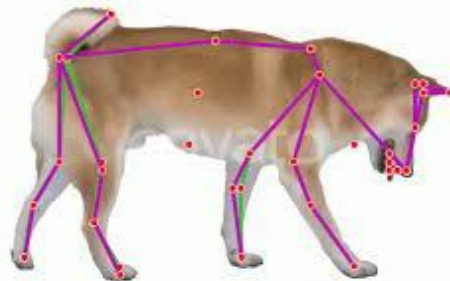


Technical Hardships / Project Challenges

- Only 14 dogs total (7 CCL, 7 Normal) across 12 GB of video recorded under inconsistent conditions with varied camera distances, angles, etc. made standardization challenging.
 - We also realized how much inter-subject gait variation there was.
- SuperAnimal-Quadruped occasionally swapped left/right keypoints or produced low-confidence estimates, so we had to implement confidence thresholding.
- Initially tried using ViTPose+ with AnimalPose 10K keypoint schema
 - Had poor joint landmarks / coverage in the lower leg
- Running pose estimation across ~450 candidate videos on CPU was prohibitively slow (hours). We used a pre-filter to reduce the set to only usable lateral clips before running the pose model.



Sample Demo on Stock Video



Project Demo: Meet NaeNae



Nae Nae's Results



=== [4/4] Result ===

Video: trial_sample.MOV

Model: random_forest

Pose coverage: 99%

L stifle (mean): 111.5°

R stifle (mean): 143.3°

|L-R| stifle: 42.8°

P(CCL): 0.594

Prediction: CCL (injured)

Confidence: 59.4%



Conclusions

- Our pipeline achieved 94.6% clip accuracy and 100% dog-level accuracy on the held-out dogs, successfully distinguishing CCL-injured from healthy using just the smartphone captured videos.
- The pipeline, though it requires much more polish and data than we can afford to give, is promising as one part of a larger clinical gait assessment.

